Google Developer Groups

Responsibilities and Pre-Interview Tasks

A Note for First-Year Students

Seeing these tasks for the first time might seem intimidating, and that's perfectly okay! Your main goal isn't to create a flawless project. It's to dive in, explore, and get familiar with the exciting concepts of development :)

Tackle these challenges on your own and see how far you can get. Remember, your effort and curiosity are what truly count. We value your willingness to learn far more than a perfect solution.

All the best and happy building!

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Technical Domains

1. AI/ML

RESPONSIBILITIES

- Develop hands-on familiarity with AI/ML tools, frameworks, and emerging technologies.
- Build and experiment with models, prototypes, and beginner-friendly AI applications.
- Contribute to technical workshops, demos, and curated learning resources.

Pre-Interview Tasks

First Year

Prerequisite Tasks

Learn the basics behind ML statistics and the mathematical terminology you will need:

- Data, mean, mode, deviations, kurtosis, basic statistics, and matrices.
- The terms precision, recall, and F1 score.
- Basic vector transformations and number systems.

No project is required for this prerequisite section.

House Price Predictor (Linear Regression)

1. Build a very basic house price prediction model using Linear Regression.
2. Use a simple dataset containing house size (area) and house price.
3. Explore the data and understand how house size affects house prices.
4. Train a Linear Regression model to predict house prices based on house size.
5. Evaluate the model by comparing predicted prices with actual prices.
6. Plot a simple graph showing house sizes, actual prices, and the prediction line.
7. **Optional:** Create a basic interface where a user enters the house area and gets a predicted price.

Second & Third Year

Project Track A: Choose One Project

Choose one of the two projects below and showcase it as a working app with a GitHub repository.

Option A: Insurance Cost Predictor

Build a regression model on the Medical Cost Personal Dataset (age, sex, BMI, children, smoker, region).

1. Explore the data and preprocess it (encoding, skewed values).
2. Compare at least three models, such as Linear Regression, Random Forest, and Gradient Boosting.
3. Evaluate with MAE, RMSE, and R^2 using 5-fold cross-validation.
4. Put the best model in a Streamlit or Gradio app where a user enters their details and gets a predicted charge.

Option B: Review Sentiment Analyzer

Classify movie or product reviews as positive or negative using two different approaches.

1. Build a baseline with TF-IDF and Logistic Regression.
2. Compare it with a pretrained transformer from Hugging Face.
3. Report accuracy, precision, recall, F1, and a confusion matrix.
4. Build a web app that takes any review and shows the sentiment and confidence from both models.

Deliverables

1. A GitHub repository with organized folders (data, notebooks, app, `requirements.txt`).
2. A README with setup steps, screenshots, and a results table.
3. A working demo, either local or hosted on Streamlit Cloud or Hugging Face Spaces.

Project Track B: Choose One Project

Choose one of the two projects below and build it end to end, from data to a deployed and documented system.

Option 1: RAG Chatbot over Documents

Build a retrieval-augmented chatbot over a set of PDFs such as a syllabus, policy documents, or research papers.

1. Ingest and chunk the documents, then build retrieval using embeddings and a vector store.
2. Generate answers with citations to the source passages.
3. Test it on at least 20 questions, including some the documents cannot answer.

Option 2: Fine-Tuned Transformer vs. Classical Baseline

Fine-tune a small transformer such as DistilBERT on a real text classification task, then:

1. Build a classical ML baseline on the same data.
2. Compare both models on accuracy, F1, training time, and inference cost.

Deliverables

1. A GitHub repository with a clear structure, reproducible setup, and a detailed README.
2. A deployed demo, or a Dockerized app with a FastAPI backend and a UI.
3. An evaluation report in the repository covering metrics, error analysis, failure cases, and bias or ethical risks.
4. A 3-minute demo video (optional).

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2. Web Development

RESPONSIBILITIES

- Develop and maintain web applications using modern web technologies.
- Work on frontend and backend features for club projects and the official website.
- Contribute to web development workshops, demos, and collaborative projects.

Pre-Interview Tasks

First Year

Interactive Campus Explorer

Objective: Build a responsive website that helps students explore campus clubs, events, or student resources.

Requirements:

1. Create a visually appealing, responsive webpage that displays information through cards or sections.
2. Implement JavaScript-based search or filtering to help users find relevant information.
3. Include at least one interactive feature, such as expandable event details, category filters, or a registration form with input validation.
4. Ensure the website works well on both mobile and desktop screens.

Second & Third Year

Public Data Explorer

Objective: Build a web application that retrieves and presents information from a public API or open dataset.

Requirements:

1. Choose a suitable public API or dataset and build a user-friendly interface around it.
2. Fetch and display data dynamically rather than relying on hardcoded information.
3. Implement search, filtering, or sorting to help users explore the data.
4. Handle loading states, empty results, and API errors.
5. Ensure the application is responsive and presents information clearly.

Optional: Build a Chrome extension that solves a real-world problem.

Note for third-year students: You are expected to extend the project with an additional feature or technical enhancement of your choice, demonstrating greater complexity and independent problem-solving.

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3. App Development

RESPONSIBILITIES

- Build and test mobile applications using Kotlin, Java, or Flutter.
- Work on app features, UI implementation, debugging, and deployment.
- Contribute to mobile development workshops and project showcases.

Pre-Interview Tasks

First Year

Daily Habit Tracker

Objective: Build a simple mobile app using Flutter (preferred), Java, or Kotlin that lets users add, view, complete, and delete daily habits, with a counter showing completed habits.

Requirements:

- Demonstrate programming fundamentals, including variables, conditionals, functions, lists, user input, basic UI development, and basic OOP concepts.

Optional: Store habits locally so they persist after the app restarts.

Second & Third Year

Quiz App

Objective: Build a mobile app using Flutter (preferred), Java, or Kotlin that lets users choose from at least two quiz categories, answer multiple-choice questions, and view their final score.

Requirements:

- Demonstrate advanced OOP concepts, data modeling, state management, and separation of UI from business logic.

Optional: Store questions or previous scores using SQLite or Firebase.

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4. Web3

RESPONSIBILITIES

- Explore blockchain technologies, smart contracts, and decentralized applications.
- Build basic Web3 prototypes and experiment with tools such as Solidity.
- Contribute to technical demos, workshops, and discussions on decentralized technologies.

Pre-Interview Tasks

Don't aim for perfection. Your effort and curiosity are what truly count.

Who Completes What

- **First Year:** Any 1 of the 3 tasks.
- **Second & Third Year:** Any 2 of the 3 tasks.

Task 1: Decentralized Crowdfunding dApp (FundChain)

Objective: Users create campaigns and contribute testnet ETH. The creator withdraws only if the goal is met by the deadline; otherwise, contributors get refunds.

1. **Contract:** Solidity on a testnet (e.g., Sepolia) with a Campaign struct and createCampaign(), contribute() (payable), withdrawFunds(), and refund().
2. **Frontend:** Wallet connection, a list of campaigns with progress and time left, and the right action (withdraw or refund) shown for each campaign state.

Task 2: Blog on Account Abstraction (ERC-4337)

Objective: Write a technical blog post on how Account Abstraction can make Web3 easier and safer.

1. Cover EOAs vs. smart wallets, how ERC-4337 works at a high level, key benefits (gasless transactions, social recovery, session keys, batching), and its limitations.
2. Include real-world examples (e.g., Safe, Argent, Coinbase Smart Wallet) and at least one diagram you made yourself.

Deliverable: A blog of about 1,000–1,500 words, with sources cited.

Task 3: Decentralized Voting App (dVote)

Objective: Users create proposals and vote with their wallets, one vote per wallet per proposal.

1. **Contract:** A Proposal struct with createProposal(), vote(), and getAllProposals(). Block double voting and votes after the deadline.
2. **Frontend:** Wallet connection, live vote counts, Active/Closed status, whether you have voted, and clear error messages.

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5. Cloud

RESPONSIBILITIES

- Work with cloud platforms and explore core cloud computing services like GCP.
- Deploy applications and experiment with cloud infrastructure and services.
- Contribute to cloud-based projects, practical demos, and workshop activities.

Pre-Interview Tasks

First Year

Cloud Computing Fundamentals

Prepare a PPT covering the fundamental concepts of cloud computing. Your presentation should cover:

- Cloud computing fundamentals.
- Differences between IaaS, PaaS, SaaS, and other cloud service models.
- Benefits of cloud computing over traditional on-premises systems.
- Types of cloud services and service providers (AWS, Azure, GCP, etc.).
- Real-world use cases and examples (at least 2).

Note: You can also mention your previous experience for additional consideration.

Second & Third Year

Cloud Deployment Task

Deploy and demonstrate any one of your projects, or a real-world application, using a cloud service (GCP preferred). Implement at least **one** of the following integrations:

- **Database integration:** Integrate a cloud database, such as Firestore, Cloud SQL, or DynamoDB, and demonstrate CRUD operations.
- **Cloud security:** Implement basic security measures, such as appropriate IAM roles, secure API key management, or firewall rules.
- **Environment configuration:** Configure autoscaling or relevant environment settings and demonstrate the configuration.

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6. Cybersecurity

RESPONSIBILITIES

- Explore network security, secure coding, and ethical hacking techniques.
- Participate in vulnerability assessments, CTF challenges, and practical security exercises.
- Take part in hands-on bug bounties and collaborative bug hunting.
- Develop security demos and contribute to cybersecurity workshops and awareness initiatives.

Pre-Interview Tasks

First Year

- Prepare a PPT on any cybersecurity concept or vulnerability, with a deep dive into its core issue.
- Read about the OWASP Top 10.

Second & Third Year

- Practice CTF challenges.
- Learn about CVE IDs and the core issues behind the OWASP Top 10.

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Non-Technical Domains

1. Finance

RESPONSIBILITIES

- Maintain expense records, budget sheets, and financial documentation for club activities.
- Research vendor pricing and compare costs to support resource planning.
- Track event expenses and ensure financial records remain accurate and organized.

Pre-Interview Tasks

1. **Budget Planning:** Prepare a detailed budget for a 24-hour hackathon with 100 participants, covering all major expenses and estimating the total cost (in ₹).
2. **Budget Optimization:** Revise your proposed budget to reduce total expenditure by 40% while maintaining the quality and essential requirements of the hackathon.
3. **Fund Allocation:** Suppose an additional sponsorship of ₹30,000 is secured. Explain how you would allocate these funds to maximize the event's success and the participant experience.

2. Sponsorship

RESPONSIBILITIES

- Research potential sponsors and identify relevant partnership opportunities.
- Develop sponsorship decks, proposals, and outreach material.
- Engage with prospective sponsors and support partnership communication and follow-ups.

Pre-Interview Tasks

1. Make a list of 5 potential sponsors, with reasons for choosing each.
2. Draft a professional email to approach a potential sponsor.
3. Who would be your dream sponsor for a GDGoC event? Prepare a formal pitch to present to them.
4. Find 5 local brands or small businesses as potential sponsors, and list unique deliverables you would offer for the amount secured from each.

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3. Publicity & Outreach

RESPONSIBILITIES

- Drive event promotion through publicity, campus networks, and online channels.
- Build connections with other GDGs and coordinate cross-promotional initiatives.
- Develop outreach strategies to increase event visibility and community participation.

Pre-Interview Tasks

1. Suggest 5 creative initiatives you would introduce to make the Publicity & Outreach workflow smoother and improve coordination between the two campuses.
2. Suggest 3 unconventional ways to increase GDGoC's outreach and visibility among students of other colleges.
3. Suggest 5 creative desk activities that could attract students to the GDGoC desk and make them curious about the community.
4. You have to publicize a GDGoC event across campus in one day, with a team of 4 and a limited budget. Create a plan of action for the day.
5. Suggest 3 publicity ideas that would make GDGoC stand out from other college clubs during recruitment or major events.
6. GDGoC is organizing an event, but registrations are much lower than expected 3 days before it. What would you do to increase registrations?

4. Execution

RESPONSIBILITIES

- Manage event logistics, including venue setup, permissions, equipment, registrations, and attendee coordination.
- Coordinate with different domains to ensure event requirements and timelines are met.
- Handle on-ground operations and resolve logistical challenges during events.

Pre-Interview Tasks

1. Write the event flow for an App Development workshop.
2. How would you manage a crowd of 200 attendees when the venue can accommodate only 100?
3. What would you do if a technical failure occurred during the event?

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5. Content

RESPONSIBILITIES

- Create engaging content for social media, newsletters, blogs, and event communications.
- Research technical and non-technical topics and translate them into clear, audience-friendly content.
- Develop creative concepts, maintain content schedules, and contribute to the communication strategy.

Pre-Interview Tasks

Before the interview, explore GDGoC VIT Pune's Instagram and LinkedIn pages to understand the kind of content we put out. Using your observations, complete the tasks below in two styles:

- **GDGoC approach:** Follow the tone, writing style, and content style you observe on our platforms.
- **Your approach:** Tackle the same tasks in your own creative style, reflecting the kind of content you would like to create for GDGoC.

Tasks

1. Write a concise newsletter summary section for a recent GDGoC workshop, highlighting its key takeaways and relevance to students.
2. Write a LinkedIn post recapping a GDGoC event and highlighting its key outcomes.
3. Propose a recurring content series for the GDGoC Instagram, with a name and three post ideas.
4. Turn a technical concept of your choice into a catchy headline and a short, beginner-friendly social media caption.

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6. Multimedia

RESPONSIBILITIES

- Design posters, social media creatives, presentations, and promotional materials.
- Create and edit reels, event highlights, and other video content.
- Handle event photography and videography while maintaining consistency across the chapter's visual identity.

Pre-Interview Tasks

Choose any one sub-domain: Graphic Design, Video Editing, or Photography/Videography.

Graphic Design

- Create a seamless 1×3 Instagram grid for an imaginary GDGoC event, including all relevant event details.
- Create an Instagram story for the event above or for any festival.

Two variants: For both tasks, create one version in your favorite aesthetic or design trend, showcasing your creativity, and another following Google's professional branding and visual identity.

Video Editing / Photography / Videography

- Create a short reel showcasing your video editing skills and creativity.
- Showcase your photography or videography skills through a collection of photographs or video clips.

Two variants: For your chosen task, create one version inspired by the video style and visual storytelling of the official Google Developer Groups (GDG) social media pages, and another reflecting your own creative vision and personal style.

Note: A portfolio showcasing your best previous work, tools, and skillset is mandatory for all sub-domains.

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7. Aesthetics

RESPONSIBILITIES

- Conceptualize and create event decor, physical installations, and thematic displays.
- Design DIY artifacts and source materials that align with event themes and branding.
- Execute venue styling and coordinate with other domains to create cohesive event environments.

Pre-Interview Tasks

DIY Tasks

- **Color Coordination:** Pick 3–4 colors for a GDGoC event and create a physical color combination using objects, paper, or materials around you. Explain your choices.
- **Creative Setup:** Visit a classroom or seminar hall in your college and suggest how you would decorate it for a GDGoC workshop. Include a rough hand-drawn layout of your plan.
- **Campus Aesthetics:** Identify one spot on your college campus that could look more attractive during a GDGoC event. Take a photograph of the spot and suggest 3 practical ways to improve it within a limited budget.
- **3D Letter Creation:** Create a 3D letter “G”, 12 cm tall, using easily available materials. Make it creative, neat, and suitable for a GDGoC college club.

Written Tasks

- **Event Theme:** Choose one theme for a GDGoC college event and explain your choice with a real-life example or reference that inspired you.
- **Decoration Ideas:** Suggest 3 decoration ideas for a GDGoC event using materials available at home or in college. Mention the materials and the approximate cost of each.